

WLM Foundation Series
Appendix: Unified Glossary and Axiom–Theorem Dependency Matrix
v3.0 • April 2026 • Final Edition • Covers Papers 0–5

Wujie Gu
World-Layer Model (WLM) Series

Copyright © 2026 Wujie Gu. NCND License.

1. Unified Glossary

All terms used across Papers 0–5 of the WLM Foundation Series. Terms are listed in order of conceptual depth: foundational equivalence relations first, then each generative layer in order. Every paper inherits these definitions without restatement.

Term	Symbol	Definition	First Defined
Observational Equivalence	$\sim_o$	$x \sim_o y \Leftrightarrow \forall O \in \mathcal{O}, O(x) = O(y)$. Equivalence relation on X induced by the family of observations.	Paper 1, Sec 0.1
Symmetry Interval	I_0	The equivalence class $[0]_{\sim_o} = \{x : x \sim_o 0\}$. The symmetry phase of the generative system.	Paper 1, Def 0.1
Distinguishability Valuation	$v(\cdot)$	Map $v: X \rightarrow (V, \leq)$ with $v(x)=v(y) \Leftrightarrow x \sim_o y$, $v(0)$ minimal. Realizes $\ x-0\ >0 \Leftrightarrow v(x)>v(0)$.	Paper 1, Def 0.4
Perfect Symmetry State	$\text{Sym}_p^{\text{erf}}$	$\{x \in X : \forall y, x \sim_o y\}$. Equals I_0 . The unique fixed-point class of 0.	Paper 1, Def 1.1
Symmetric Candidate	Sym^{cand}	$\{x : \forall G \in \mathcal{G}_{\{I_0\}}, G(x) \sim_o x\}$. Equals I_0 (Paper 1 Thm 2.1).	Paper 1, Def 2.1
Minimal Offset	ε	Unique (up to $\sim_o$) first distinguishable deviation from 0. $\inf\{\ x-0\ : x \not\sim_o 0\}$.	Paper 0 / Paper 1, Def 0.5
Orientation Label	$\theta(\cdot)$	Primitive orientation carried by $x \not\sim_o 0$, with $x \sim_o y \Rightarrow \theta(x)=\theta(y)$. $\theta(0)$ undefined.	Paper 2, Def 1.1
Dual Offsets	$\pm\varepsilon$	Unique oriented pair $\{+\varepsilon, -\varepsilon\}$ forced by D0 once ε exists. Binary orientation space at $v(\varepsilon)$.	Paper 0 Thm 3 / Paper 2
Generative Relations	$\mathcal{G}$	Family of total functions $G: X \rightarrow X$, closed under composition by R1.	Paper 1, Sec 0.2
I_0-Preserving Generators	$\mathcal{G}_{\{I_0\}}$	$\{G \in \mathcal{G} : G(I_0) \subseteq I_0\}$. Subfamily that maps the symmetry interval into itself.	Paper 1, Def 0.2
Generative Chain	$x_0, x_1, \dots$	Sequence with $x_0=0$ and $x_{n+1}=G_n(x_n)$ for $G_n \in \mathcal{G}$. Identical definition in Papers 0 and 1.	Paper 0 / Paper 1, Def 0.3

Recursive Generator	f (recursive)	Generator $f \in \mathcal{G}$ whose iterates f^n are defined for all $n \geq 1$ via R1.	Paper 3, Def 1.1
Invariant Structure	S	Set $S \subseteq X$ with $f(S) \subseteq S$ for some recursive f . Periodic cycle or compact closure.	Paper 4, Def 1.1
Appearance	$\Sigma = P(S)$	Image of invariant structure S under projection P guaranteed by P0. The phenomenal world-layer.	Paper 5, Def 1.1
World-Layer	—	Observational system with phenomenal content: $\exists O$ with $O(x) \neq O(y)$ for some $x \neq y$. Requires appearance.	Paper 5
Trivial System	G_{triv}	System with single state, identity mapping, no non-trivial invariants. Standard of observational emptiness.	Paper 0 Thm 6

2. Axiom Summary

The six axioms of the WLM Foundation Series. Each axiom is stated once in the paper where it first becomes structurally necessary; all subsequent papers inherit it.

Axiom	Name	Role in Chain	First Used
S0	Symmetry Closure	0 is a fixed point; forces departure via ε	Paper 0 / Paper 1
D0(a)	Primitive Orientation	Any $x \neq_0 0$ carries orientation label $\theta(x)$	Paper 2
D0(b)	Orientation Reversal	Ensures reversal generator F exists; makes $\pm \varepsilon$ operationally realizable	Paper 2
R1	Composability	Closes generators under composition; makes recursion possible	Papers 0–3
R2	Orientation Preservation	Directional flips persist under all iterates; blocks directionality washout	Papers 0, 2–4
V-WF	Well-Founded Valuation	Every nonempty subset above $v(0)$ has a minimal element; ensures ε exists and is unique	Paper 1
P0	Projection Existence	Any system with stable invariants admits at least one projection to an observational layer	Papers 0, 4–5

Reading note. *V-WF is the workhorse of Paper 1 and is used only there. S0 and D0 are used early and tapering off by Paper 3. R1 and R2 become central from Paper 2 onward. P0 appears only in Papers 0 and 4–5, isolating the appearance step. The clean separation of axiom domains reflects the layered architecture of the chain.*

3. Axiom–Theorem Dependency Matrix

Rows are theorems and lemmas from Papers 0–5, grouped by paper. Columns are the six axioms plus four frequently-cited prior results: P1 T3.3 = Paper 1 Theorem 3.3 (ϵ unskippable); Lem4c = Paper 0 Lemma 4c (persistent $\pm\epsilon$ alternation); B-W = Bolzano-Weierstrass theorem.

✓ = direct dependence. · = not used. Header rows (shaded) separate papers.

Theorem / Lemma	S0	D0(a)	D0(b)	R1	R2	V-WF	P0	P1 T3.3	Lem4c	B-W
Paper 0										
Thm 1 (0 is fixed)	✓	·	·	·	·	·	·	·	·	·
Thm 3 (Forced Duality)	·	✓	✓	·	·	·	·	·	·	·
Lemma 4c (Alternation)	·	✓	·	✓	✓	·	·	·	·	·
Thm 5 (Recursion $\rightarrow$ Invariants)	·	·	·	✓	·	·	·	·	✓	✓
Thm 6 (Structure $\rightarrow$ Appearance)	·	·	·	·	·	·	✓	·	·	·
Paper 1										
Thm 1.1–1.2 ($\text{Sym}_p^{\text{erf}} = I_0$)	✓	·	·	·	·	·	·	·	·	·
Thm 2.1–2.2 ($\text{Sym}^{\text{cand}} = I_0$)	✓	·	·	·	·	·	·	·	·	·
Thm 3.1–3.4 (ϵ necessity & uniqueness)	·	·	·	·	·	✓	·	·	·	·
Thm 3.3 (ϵ unskippable)	·	·	·	·	·	✓	·	·	·	·
Thm 4.1–4.2 (collapse modes)	·	·	·	·	·	✓	·	·	·	·
Lemma 4.3 (latent directionality)	·	✓	·	·	·	✓	·	·	·	·
Paper 2										
Thm 1.1 (Orientation Minimality)	·	✓	·	·	·	·	·	✓	·	·
Thm 1.2 (Orientation Non-Degeneracy)	·	✓	·	·	·	·	·	✓	·	·
Thm 2.1 (Uniqueness of $\pm\epsilon$)	·	✓	✓	·	·	·	·	✓	·	·
Thm 2.2 (No Third Direction)	·	✓	·	·	·	·	·	✓	·	·
Thm 3.1 (Directionless Collapse)	·	✓	·	·	·	·	·	·	✓	·
Thm 3.2 (Asymmetry Collapse)	·	✓	·	·	·	·	·	✓	✓	·
Thm 4.1 (Duality Preservation)	·	·	·	✓	✓	·	·	·	·	·
Thm 4.2 (Duality = Unique Generator)	·	·	·	✓	✓	·	·	·	✓	·
Thm 4.3 (Duality Unskippable)	·	✓	·	✓	✓	·	·	·	✓	·
Paper 3										
Thm 1.1 (Minimal Recursion)	·	·	·	✓	✓	·	·	·	✓	·
Thm 1.2 (Recursion Irreducibility)	·	·	·	✓	·	·	·	✓	✓	·
Lemma 1.3 (Recursion as Engine)	·	·	·	✓	·	·	·	·	✓	✓
Thm 2.1 (Duality Requires Recursion)	·	·	·	✓	✓	·	·	·	✓	·
Thm 2.2 (Recursion Uniqueness)	·	·	·	✓	·	·	·	·	✓	·
Thm 3.1 (Static Collapse)	·	·	·	✓	·	·	·	·	·	·
Thm 3.2 (Divergent Collapse)	·	·	·	✓	·	·	·	·	·	✓
Thm 4.1 (Invariant Generation)	·	·	·	✓	✓	·	·	·	✓	✓

Thm 4.2 (No Structure w/o Recursion)	·	·	·	✓	✓	·	·	·	·	·
Thm 4.3 (Recursion Unskippable)	·	·	·	✓	✓	·	·	·	✓	·
Paper 4										
Thm 1.1 (Minimal Structure)	·	·	·	✓	✓	·	·	·	✓	✓
Thm 1.2 (Structure Irreducibility)	·	·	·	✓	·	·	·	·	·	·
Thm 2.1 (Recursion Generates Structure)	·	·	·	✓	·	·	·	·	✓	✓
Thm 2.2 (Structure Uniqueness)	·	·	·	·	·	·	·	✓	·	·
Thm 3.1 (Oscillation Collapse)	·	·	·	✓	✓	·	·	·	✓	·
Thm 3.2 (Divergent Collapse)	·	·	·	✓	·	·	·	·	·	✓
Thm 4.1 (Invariant Stability)	·	·	·	✓	·	·	·	·	·	·
Thm 4.2 (No Appearance w/o Structure)	·	·	·	·	·	·	✓	·	·	·
Thm 4.3 (Structure Unskippable)	·	·	·	✓	✓	·	✓	·	·	·
Paper 5										
Thm 1.1 (Minimal Appearance)	·	·	·	·	·	✓	✓	·	·	·
Thm 1.2 (Appearance Irreducibility)	·	·	·	✓	✓	·	✓	✓	·	·
Thm 2.1 (Structure Generates Appearance)	·	·	·	·	·	·	✓	·	·	·
Thm 2.2 (Appearance Uniqueness)	·	·	·	·	·	·	✓	·	·	·
Thm 3.1 (Pre-Appearance Collapse)	·	·	·	·	·	·	✓	·	·	·
Thm 3.2 (Projection Failure Collapse)	·	·	·	·	·	·	✓	·	·	·
Thm 4.1 (Appearance Stability)	·	·	·	✓	·	·	✓	·	·	·
Thm 4.2 (No World w/o Appearance)	·	·	·	·	·	·	✓	·	·	·
Thm 4.3 (Appearance Unskippable)	·	·	·	·	·	·	✓	·	·	·

How to read the matrix. Reading down a column shows all results that depend on a given axiom. Reading across a row shows all dependencies of a given theorem. The matrix confirms the layered architecture: each paper adds exactly one new forced layer while depending only on prior results. R1 and R2 together drive Papers 2–4. P0 alone drives Paper 5.

4. Unskippability Results — Confirmed Partial Results for Conjecture U1

Conjecture U1 asserts that the chain $0 \rightarrow \varepsilon \rightarrow \{\pm\varepsilon\} \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ is the unique forced generative sequence with no alternative paths or shortcuts. The following five unskippability theorems are the confirmed partial results. Each result adds one verified link to the chain.

Paper	Layer	Unskippable Result	Theorem
-------	-------	--------------------	---------

1	ε	Any chain reaching non-zero states must pass through ε	Thm 3.3
2	$\pm\varepsilon$	Any chain generating non-trivial recursion must pass through $\{\pm\varepsilon\}$	Thm 4.3
3	Recursion	Any chain generating non-trivial invariant structure must pass through recursion	Thm 4.3
4	Structure	Any chain reaching appearance must pass through invariant structure	Thm 4.3
5	Appearance	Any chain realizing a world-layer must pass through appearance	Thm 4.3

The five partial results together confirm that the chain cannot be shortened: no link can be skipped. The full proof of chain uniqueness — that no alternative chain of the same length produces equivalent results via different intermediate steps — remains the central open problem of the WLM Foundation Series.

5. Series File Structure

WLM-Foundation-Series/	
— Paper0.docx	The Generative Beginning of a Universe (v3.1)
— Paper1.docx	The Stability and Uniqueness of 0 (v2.1)
— Paper2.docx	The Uniqueness and Unskippability of $\pm\varepsilon$ (v1.0)
— Paper3.docx	The Uniqueness and Unskippability of Recursion (v1.0)
— Paper4.docx	The Uniqueness and Unskippability of Invariant Structure (v1.0)
— Paper5.docx	The Uniqueness and Unskippability of Appearance (v1.0)
— Foundation_Glossary_Matrix_v3.0.docx	← this file (series closed)

The WLM Foundation Series is now complete. All six papers (Papers 0–5) and this appendix are the canonical reference set. Future work in the WLM series will build on this foundation without modifying it.

License

Copyright © 2026 Wujie Gu. All rights reserved. NCND License.

Contact: Wujie Gu • OSF: osf.io/rq5vj • ResearchGate